

INCLUDES

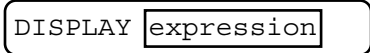
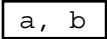
- ✓ Course framework
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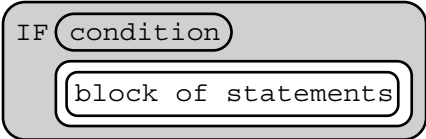
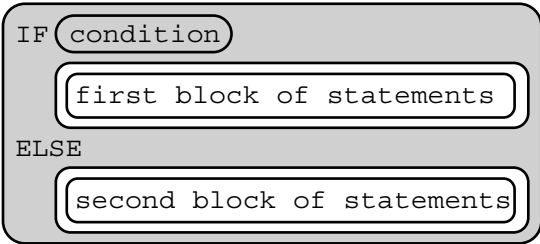
AP[®] Computer Science Principles

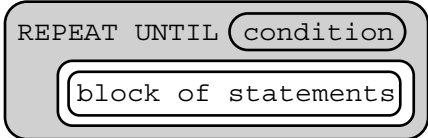
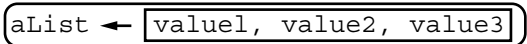
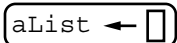
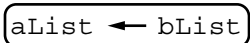
COURSE AND EXAM DESCRIPTION

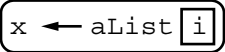
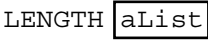
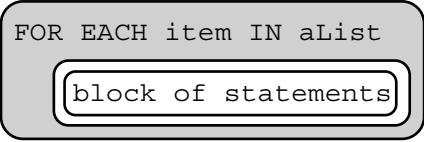
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Fall 2023

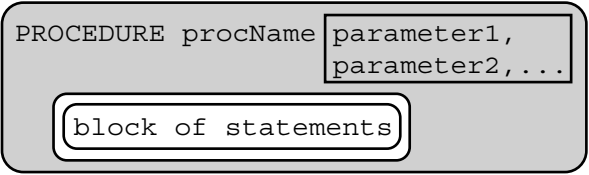
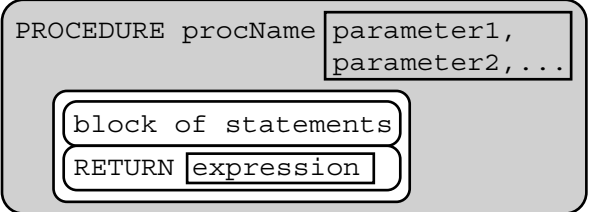
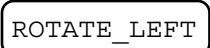
Exam Reference Sheet

Instruction	Explanation
Assignment, Display, and Input	
Text: <code>a ← expression</code> Block: 	Evaluates <code>expression</code> and then assigns a copy of the result to the variable <code>a</code> .
Text: <code>DISPLAY(expression)</code> Block: 	Displays the value of <code>expression</code> , followed by a space.
Text: <code>INPUT()</code> Block: <code>INPUT</code>	Accepts a value from the user and returns the input value.
Arithmetic Operators and Numeric Procedures	
Text and Block: <code>a + b</code> <code>a - b</code> <code>a * b</code> <code>a / b</code>	The arithmetic operators <code>+</code>, <code>-</code>, <code>*</code>, and <code>/</code> are used to perform arithmetic on <code>a</code> and <code>b</code>. For example, <code>17 / 5</code> evaluates to <code>3.4</code>. The order of operations used in mathematics applies when evaluating expressions.
Text and Block: <code>a MOD b</code>	Evaluates to the remainder when <code>a</code> is divided by <code>b</code>. Assume that <code>a</code> is an integer greater than or equal to <code>0</code> and <code>b</code> is an integer greater than <code>0</code>. For example, <code>17 MOD 5</code> evaluates to <code>2</code>. The <code>MOD</code> operator has the same precedence as the <code>*</code> and <code>/</code> operators.
Text: <code>RANDOM(a, b)</code> Block: <code>RANDOM</code> 	Generates and returns a random integer from <code>a</code> to <code>b</code>, including <code>a</code> and <code>b</code>. Each result is equally likely to occur. For example, <code>RANDOM(1, 3)</code> could return <code>1</code>, <code>2</code>, or <code>3</code>.
Relational and Boolean Operators	
Text and Block: <code>a = b</code> <code>a ≠ b</code> <code>a > b</code> <code>a < b</code> <code>a ≥ b</code> <code>a ≤ b</code>	The relational operators <code>=</code>, <code>≠</code>, <code>></code>, <code><</code>, <code>≥</code>, and <code>≤</code> are used to test the relationship between two variables, expressions, or values. A comparison using relational operators evaluates to a Boolean value. For example, <code>a = b</code> evaluates to <code>true</code> if <code>a</code> and <code>b</code> are equal; otherwise it evaluates to <code>false</code>.

Instruction	Explanation
Relational and Boolean Operators (continued)	
Text: NOT condition Block: NOT (condition)	Evaluates to true if condition is false; otherwise evaluates to false.
Text: condition1 AND condition2 Block: (condition1) AND (condition2)	Evaluates to true if both condition1 and condition2 are true; otherwise evaluates to false.
Text: condition1 OR condition2 Block: (condition1) OR (condition2)	Evaluates to true if condition1 is true or if condition2 is true or if both condition1 and condition2 are true; otherwise evaluates to false.
Selection	
Text: IF(condition) { <block of statements> } Block: 	The code in block of statements is executed if the Boolean expression condition evaluates to true; no action is taken if condition evaluates to false.
Text: IF(condition) { <first block of statements> } ELSE { <second block of statements> } Block: 	The code in first block of statements is executed if the Boolean expression condition evaluates to true; otherwise the code in second block of statements is executed.

Instruction	Explanation
Iteration	
Text: <pre>REPEAT n TIMES { <block of statements> }</pre> Block: 	The code in block of statements is executed n times.
Text: <pre>REPEAT UNTIL(condition) { <block of statements> }</pre> Block: 	The code in block of statements is repeated until the Boolean expression condition evaluates to true.
List Operations	
For all list operations, if a list index is less than 1 or greater than the length of the list, an error message is produced and the program terminates.	
Text: <pre>aList ← [value1, value2, value3, ...]</pre> Block: 	Creates a new list that contains the values value1, value2, value3, and ... at indices 1, 2, 3, and ... respectively and assigns it to aList.
Text: <pre>aList ← []</pre> Block: 	Creates an empty list and assigns it to aList.
Text: <pre>aList ← bList</pre> Block: 	Assigns a copy of the list bList to the list aList. For example, if bList contains [20, 40, 60], then aList will also contain [20, 40, 60] after the assignment.
Text: <pre>aList[i]</pre> Block: <pre>aList [i]</pre>	Accesses the element of aList at index i. The first element of aList is at index 1 and is accessed using the notation aList[1].

Instruction	Explanation
List Operations (continued)	
Text: <code>x ← aList[i]</code> Block: 	Assigns the value of <code>aList[i]</code> to the variable <code>x</code> .
Text: <code>aList[i] ← x</code> Block: 	Assigns the value of <code>x</code> to <code>aList[i]</code> .
Text: <code>aList[i] ← aList[j]</code> Block: 	Assigns the value of <code>aList[j]</code> to <code>aList[i]</code> .
Text: <code>INSERT(aList, i, value)</code> Block: 	Any values in <code>aList</code> at indices greater than or equal to <code>i</code> are shifted one position to the right. The length of the list is increased by 1, and <code>value</code> is placed at index <code>i</code> in <code>aList</code> .
Text: <code>APPEND(aList, value)</code> Block: 	The length of <code>aList</code> is increased by 1, and <code>value</code> is placed at the end of <code>aList</code> .
Text: <code>REMOVE(aList, i)</code> Block: 	Removes the item at index <code>i</code> in <code>aList</code> and shifts to the left any values at indices greater than <code>i</code> . The length of <code>aList</code> is decreased by 1.
Text: <code>LENGTH(aList)</code> Block: 	Evaluates to the number of elements in <code>aList</code> .
Text: <code>FOR EACH item IN aList</code> <code>{</code> <code> <block of statements></code> <code>}</code> Block: 	The variable <code>item</code> is assigned the value of each element of <code>aList</code> sequentially, in order, from the first element to the last element. The code in <code>block of statements</code> is executed once for each assignment of <code>item</code> .

Instruction	Explanation
Procedures and Procedure Calls	
Text: <pre>PROCEDURE procName(parameter1, parameter2, ...)</pre> <pre>{ <block of statements> }</pre> Block: 	Defines procName as a procedure that takes zero or more arguments. The procedure contains block of statements. The procedure procName can be called using the following notation, where arg1 is assigned to parameter1, arg2 is assigned to parameter2, etc: <pre>procName(arg1, arg2, ...)</pre>
Text: <pre>PROCEDURE procName(parameter1, parameter2, ...)</pre> <pre>{ <block of statements> RETURN(expression) }</pre> Block: 	Defines procName as a procedure that takes zero or more arguments. The procedure contains block of statements and returns the value of expression. The RETURN statement may appear at any point inside the procedure and causes an immediate return from the procedure back to the calling statement. The value returned by the procedure procName can be assigned to the variable result using the following notation: <pre>result ← procName(arg1, arg2, ...)</pre>
Text: <pre>RETURN(expression)</pre> Block: 	Returns the flow of control to the point where the procedure was called and returns the value of expression.
Robot	
If the robot attempts to move to a square that is not open or is beyond the edge of the grid, the robot will stay in its current location and the program will terminate.	
Text: <pre>MOVE_FORWARD ()</pre> Block: 	The robot moves one square forward in the direction it is facing.
Text: <pre>ROTATE_LEFT ()</pre> Block: 	The robot rotates in place 90 degrees counterclockwise (i.e., makes an in-place left turn).

Instruction	Explanation
Robot	
Text: ROTATE_RIGHT() Block: ROTATE_RIGHT	The robot rotates in place 90 degrees clockwise (i.e., makes an in-place right turn).
Text: CAN_MOVE(direction) Block: CAN_MOVE direction	Evaluates to <code>true</code> if there is an open square one square in the direction relative to where the robot is facing; otherwise evaluates to <code>false</code> . The value of <code>direction</code> can be <code>left</code> , <code>right</code> , <code>forward</code> , or <code>backward</code> .

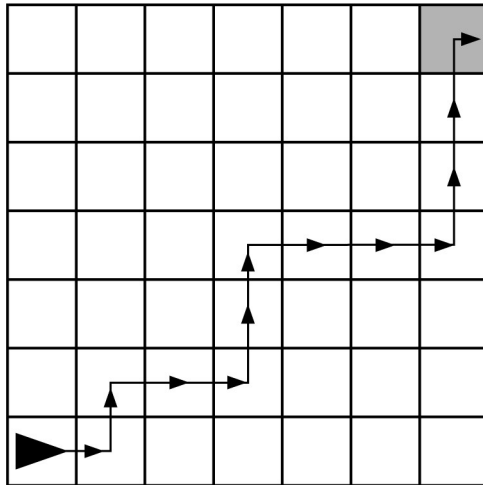
Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and AP Computer Science Principles Exam and serve as examples of the types of questions that appear on the exam. After the sample questions, you will find a table that shows the skill and learning objective(s) to which each question relates. The table also provides the answers to the multiple-choice questions.

Section I: Multiple Choice

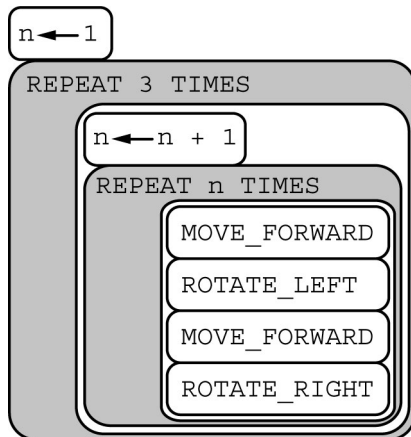
1. Which of the following best explains how data is typically assembled in packets for transmission over the Internet?
 - (A) Each packet contains data to be transmitted, along with metadata containing information used for routing the data.
 - (B) Each packet contains an encrypted version of the data to be transmitted, along with metadata containing the key needed to decrypt the data.
 - (C) Each packet contains only the metadata used to establish a direct connection so that the data can be transmitted.
 - (D) Each packet contains multiple data files bundled together, along with metadata describing how to categorize each data file.
2. Each student that enrolls at a school is assigned a unique ID number, which is stored as a binary number. The ID numbers increase sequentially by 1 with each newly enrolled student. If the ID number assigned to the last student who enrolled was the binary number 1001 0011, what binary number will be assigned to the next student who enrolls?
 - (A) 1001 0100
 - (B) 1001 0111
 - (C) 1101 0100
 - (D) 1101 0111

3. The following grid contains a robot represented as a triangle. The robot is initially facing right.

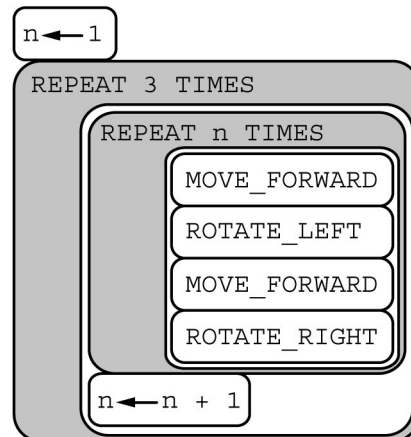


Which of the following code segments can be used to move the robot to the gray square along the path indicated by the arrows?

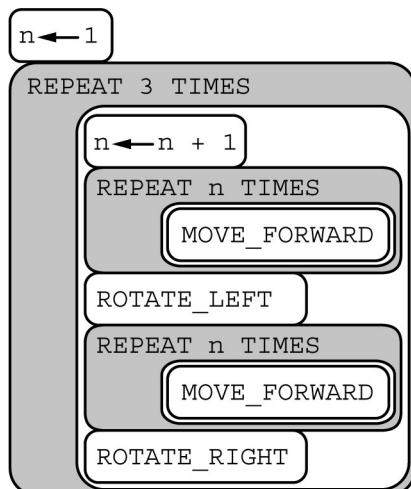
(A)



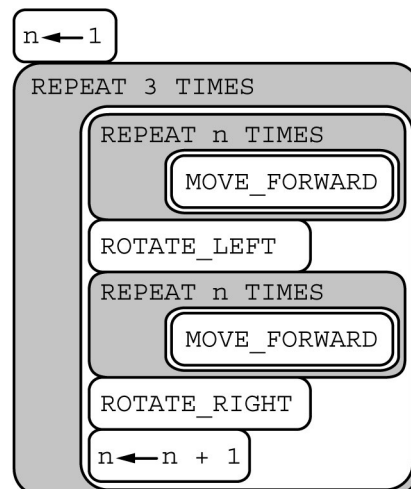
(B)



(C)



(D)



4. A user purchased a new smart home device with embedded software and connected the device to a home network. The user then registered the device with the manufacturer, setting up an account using a personal e-mail and password. Which of the following explains how a phishing attack could occur against the user of the smart home device?
- (A) A vulnerability in the device's software is exploited to gain unauthorized access to other devices on the user's home network.
 - (B) A vulnerability in the device's software is exploited to install software that reveals the user's password to an unauthorized individual.
 - (C) The user is sent an e-mail appearing to be from the manufacturer, asking the user to confirm the account password by clicking on a link in the e-mail and entering the password on the resulting page.
 - (D) The user's account is sent an overwhelming number of messages in an attempt to disrupt service on the user's home network.
5. Which of the following school policies is most likely to have a positive impact on the digital divide?
- (A) A school allows students to bring a graphing calculator from home to complete in-class mathematics assignments.
 - (B) A school allows students to bring a tablet computer to class every day to participate in graded quizzes.
 - (C) A school provides a laptop or tablet computer to all students enrolled at the school.
 - (D) A school recommends that all students purchase a computer with as much processing speed as possible so that projects run faster.
6. In a certain country, a person must be at least 16 years old to drive a car and must be at least 18 years old to vote. The variable `age` represents the age of a person as an integer.

Which of the following expressions evaluates to `true` if the person is old enough to drive but not old enough to vote, and evaluates to `false` otherwise?

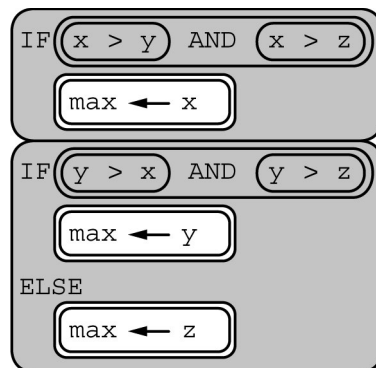
- I. $(age \geq 16) \text{ AND } (age \leq 18)$
 - II. $(age \geq 16) \text{ AND } (\text{NOT}(age \geq 18))$
 - III. $(age < 18) \text{ AND } (\text{NOT}(age < 16))$
- (A) II only
 - (B) I and II only
 - (C) I and III only
 - (D) II and III only

7. A Web site uses several strategies to prevent unauthorized individuals from accessing user accounts. Which of the following is NOT an example of multifactor authentication?
- (A) Each employee for a company is issued a USB device that contains a unique token code. To log into a company computer, an employee must insert the USB device into the computer and provide a correct password.
 - (B) After logging into an account from a new device, a user must enter a code that is sent via e-mail to the e-mail address on file with the account.
 - (C) In order to log into an account, a user must provide both a password and a fingerprint that is captured using the user's device.
 - (D) When a user enters an incorrect password more than two times in a row, the user is locked out of the account for 24 hours.
8. A list of numbers has n elements, indexed from 1 to n . The following algorithm is intended to display the number of elements in the list that have a value greater than 100. The algorithm uses the variables `count` and `position`. Steps 3 and 4 are missing.
- Step 1: Set `count` to 0 and `position` to 1.
 - Step 2: If the value of the element at index `position` is greater than 100, increase the value of `count` by 1.
 - Step 3: (missing step)
 - Step 4: (missing step)
 - Step 5: Display the value of `count`.

Which of the following could be used to replace steps 3 and 4 so that the algorithm works as intended?

- (A) Step 3: Increase the value of `position` by 1.
Step 4: Repeat steps 2 and 3 until the value of `count` is greater than 100.
- (B) Step 3: Increase the value of `position` by 1.
Step 4: Repeat steps 2 and 3 until the value of `position` is greater than n .
- (C) Step 3: Repeat step 2 until the value of `count` is greater than 100.
Step 4: Increase the value of `position` by 1.
- (D) Step 3: Repeat step 2 until the value of `position` is greater than n .
Step 4: Increase the value of `count` by 1.

9. The following code segment is intended to set `max` equal to the maximum value among the integer variables `x`, `y`, and `z`. The code segment does not work as intended in all cases.



Which of the following initial values for `x`, `y`, and `z` can be used to show that the code segment does not work as intended?

- (A) `x` = 1, `y` = 2, `z` = 3
 - (B) `x` = 1, `y` = 3, `z` = 2
 - (C) `x` = 2, `y` = 3, `z` = 1
 - (D) `x` = 3, `y` = 2, `z` = 1
10. A digital photo file contains data representing the level of red, green, and blue for each pixel in the photo. The file also contains metadata that describe the date and geographic location where the photo was taken. For which of the following goals would analyzing the metadata be more appropriate than analyzing the data?
- (A) Determining the likelihood that the photo is a picture of the sky
 - (B) Determining the likelihood that the photo was taken at a particular public event
 - (C) Determining the number of people that appear in the photo
 - (D) Determining the usability of the photo for projection onto a particular color background

11. The following procedure is intended to return the number of times the value `val` appears in the list `myList`. The procedure does not work as intended.

```
Line 1: PROCEDURE countNumOccurrences(myList, val)
Line 2: {
Line 3:   FOR EACH item IN myList
Line 4:   {
Line 5:     count ← 0
Line 6:     IF(item = val)
Line 7:     {
Line 8:       count ← count + 1
Line 9:     }
Line 10:  }
Line 11:  RETURN(count)
Line 12:}
```

Which of the following changes can be made so that the procedure will work as intended?

- (A) Changing line 6 to `IF(item = count)`
 - (B) Changing line 6 to `IF(myList[item] = val)`
 - (C) Moving the statement in line 5 so that it appears between lines 2 and 3
 - (D) Moving the statement in line 11 so that it appears between lines 9 and 10
12. A certain computer has two identical processors that are able to run in parallel. Each processor can run only one process at a time, and each process must be executed on a single processor. The following table indicates the amount of time it takes to execute each of three processes on a single processor. Assume that none of the processes are dependent on any of the other processes.

Process	Execution Time on Either Processor
X	60 seconds
Y	30 seconds
Z	50 seconds

Which of the following best approximates the minimum possible time to execute all three processes when the two processors are run in parallel?

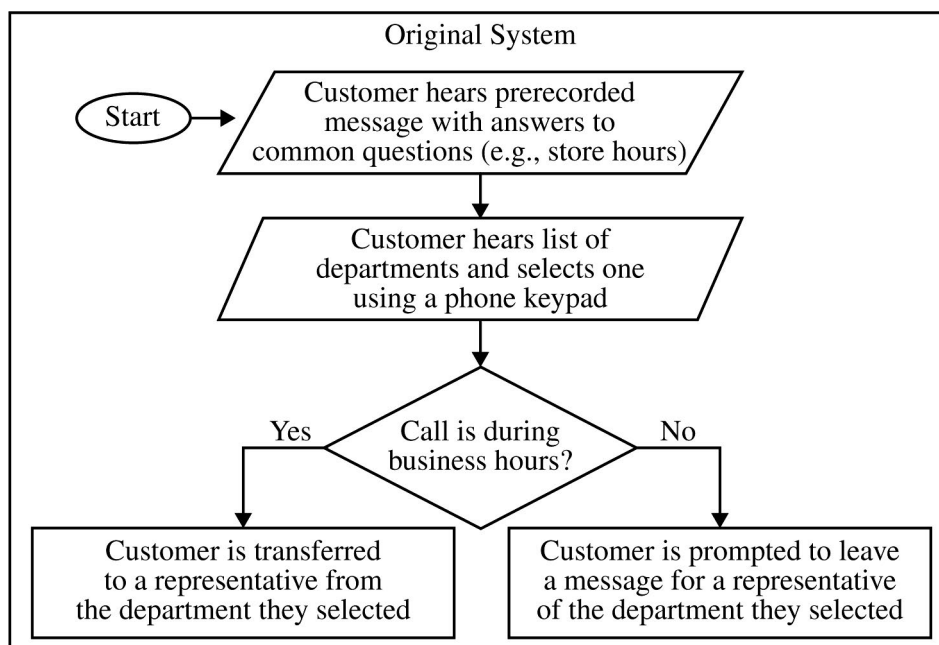
- (A) 60 seconds
- (B) 70 seconds
- (C) 80 seconds
- (D) 90 seconds

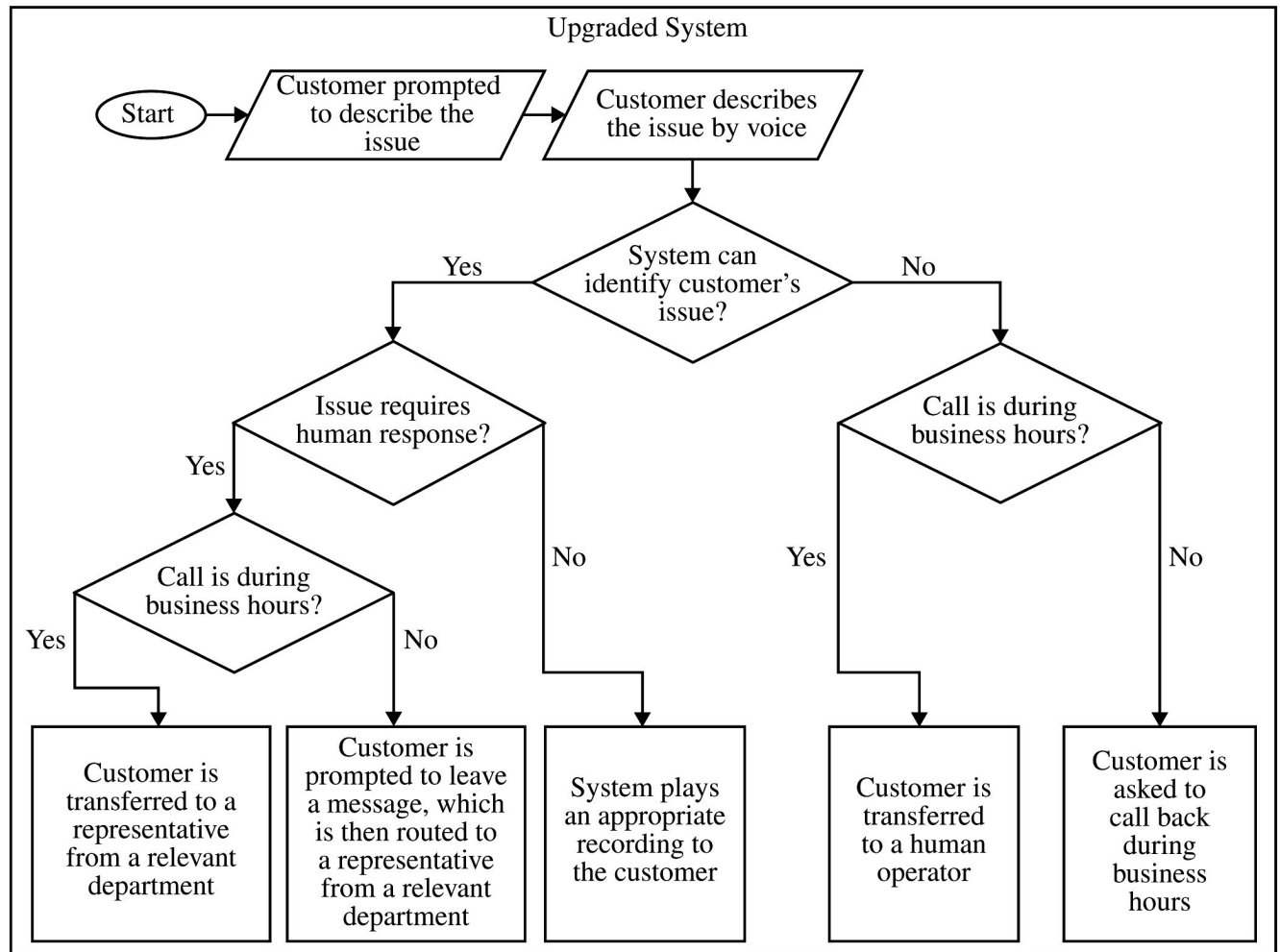
13. A sorted list of numbers contains 500 elements. Which of the following is closest to the maximum number of list elements that will be examined when performing a binary search for a value in the list?
- (A) 10
(B) 50
(C) 250
(D) 500

Questions 14–16 refer to the information below. Note: This is a representative sample of questions related to the reading passage. There will be five single-select multiple-choice questions related to a reading passage on the AP Exam.

- A chain of retail stores uses software to manage telephone calls from customers. The system was recently upgraded. Customers interacted with the original system using their phone keypad. Customers interact with the upgraded system using their voice.
- The upgraded system (but not the original system) stores all information from the calling session in a database for future reference. This includes the customer's telephone number and any information provided by the customer (name, address, order number, credit card number, etc.).
- The original system and the upgraded system are described in the following flowcharts. Each flowchart uses the following blocks.

Block	Explanation
Oval	The start of the algorithm
Parallelogram	An input or output step
Diamond	A conditional or decision step, where execution proceeds to the side labeled "Yes" if the answer to the question is yes and to the side labeled "No" if the answer to the question is no
Rectangle	The result of the algorithm





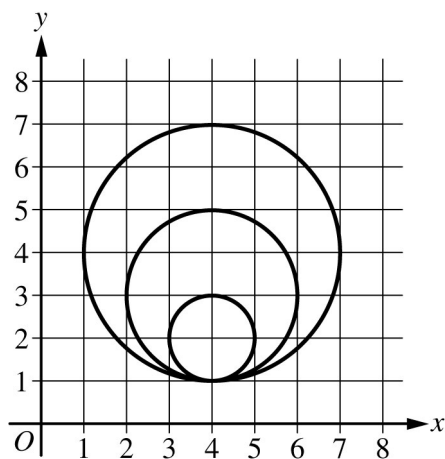
14. The upgraded system uses a directory containing additional information not supplied by the customer. The directory is used to help direct calls effectively. Which of the following is LEAST likely to be included in the directory?
- (A) A list of common issues and whether each issue requires a human representative
 - (B) A list of common keywords or phrases and a corresponding issue for each keyword or phrase
 - (C) A list of computers the company owns and the computers' corresponding IP addresses
 - (D) A list of human representatives and the corresponding department for each representative

15. Of the following potential benefits, which is LEAST likely to be provided by the upgraded system?
- (A) Human representatives will not be needed to respond to some inquiries.
 - (B) The company will be able to provide a human representative for any incoming call.
 - (C) Customers are likely to spend less time listening to information not relevant to their issue.
 - (D) Customers will be unable to mistakenly select the incorrect department for their particular issue.
16. Which of the following is the most likely data privacy concern of the upgraded system?
- (A) Customers' personal information could be compromised if an unauthorized individual gains access to the call session database.
 - (B) Storing information in the call session database makes it easy for individuals to trick the system using malicious links.
 - (C) The system design increases the chance that customers will unknowingly install malware on their devices that will share their data with unauthorized individuals.
 - (D) The system design makes it easy for unauthorized individuals to acquire customers' private encryption keys.

17. Consider the following procedure.

Procedure Call	Explanation
<code>drawCircle(xPos, yPos, rad)</code>	Draws a circle on a coordinate grid with center <code>(xPos, yPos)</code> and radius <code>rad</code>

The `drawCircle` procedure is to be used to draw the following figure on a coordinate grid.



Which of the following code segments can be used to draw the figure?

Select two answers.

(A) `x ← 4`
`y ← 1`
`r ← 0`
 REPEAT 3 TIMES
 {
`drawCircle(x, y, r)`
`r ← r + 1`
`y ← y + 1`
 }

(B) `x ← 4`
`y ← 1`
`r ← 0`
 REPEAT 3 TIMES
 {
`r ← r + 1`
`y ← y + 1`
`drawCircle(x, y, r)`
 }

(C) `x ← 4`
`y ← 4`
`r ← 3`
 REPEAT 3 TIMES
 {
`drawCircle(x, y, r)`
`y ← y - 1`
`r ← r - 1`
 }

(D) `x ← 4`
`y ← 4`
`r ← 3`
 REPEAT 3 TIMES
 {
`y ← y - 1`
`r ← r - 1`
`drawCircle(x, y, r)`
 }

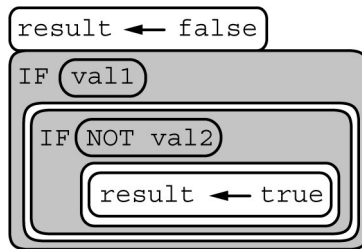
18. In the following statement, `val1`, `val2`, and `result` are Boolean variables.

```
result ← val1 AND (NOT val2)
```

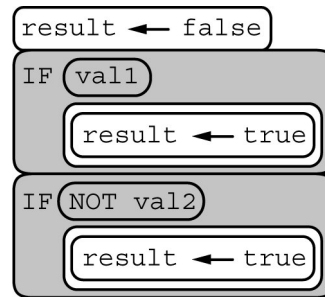
Which of the following code segments produce the same result as the statement above for all possible values of `val1` and `val2`?

Select two answers.

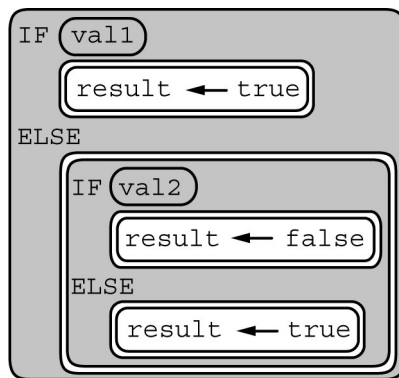
(A)



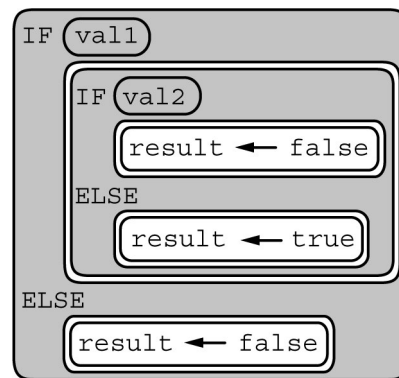
(B)



(C)



(D)



Section II: Create Performance Task: Written Responses

The Create performance task section of the end-of-course exam consists of two questions that require students to write responses that demonstrate understanding of their personal Create performance task. The two written response questions include four distinct prompts: Written Response 1, Written Response 2(a), Written Response 2(b), and Written Response 2(c). The following are sample prompts – but the specific prompts will vary across the different versions of the exam. Students will have access to their student-authored Personalized Project Reference, as long as it was submitted as final via the AP Digital Portfolio, when responding to these prompts.

Prompt Category	Sample Prompt
Written Response 1: Program Design, Function, and Purpose	Identify an expected user of your program. Describe one way your program's design meets the needs of this user.
Written Response 2(a): Algorithm Development	Consider the first iteration statement included in the Procedure section of your Personalized Project Reference. Identify the number of times the body of your iteration statement will execute. Describe a condition or error that would cause your iteration statement to not terminate and cause an infinite loop. If no such condition or error exists, explain how the loop could be modified to cause an infinite loop.
Written Response 2(b): Errors and Testing	Consider the procedure included in part (i) of the Procedure section of your Personalized Project Reference. Describe a change to your procedure that will result in a run-time error. Explain why this change will result in a run-time error.
Written Response 2(c): Data and Procedural Abstraction	Suppose you are provided with a procedure called <code>isEqual(value1, value2)</code> . The procedure returns <code>true</code> if the two parameters <code>value1</code> and <code>value2</code> are equal in value and returns <code>false</code> otherwise. Using the list you identified in the List section of your Personalized Project Reference, explain in detailed steps an algorithm that uses <code>isEqual</code> to count the number of times a certain value appears in your list. Your explanation must be detailed enough for someone else to write the program code.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective
1	A	5.A	CSN-1.C
2	A	2.B	DAT-1.C.b
3	D	2.B	AAP-2.K.a
4	C	5.E	IOC-2.C
5	C	5.C	IOC-1.C
6	D	4.B	AAP-2.F.b
7	D	5.E	IOC-2.B
8	B	2.A	AAP-2.J
9	D	4.C	CRD-2.J
10	B	5.B	DAT-2.B
11	C	4.C	CRD-2.I.b
12	C	1.D	CSN-2.A.b
13	A	1.D	AAP-2.P.a
14	C	3.A	CRD-2.C
15	B	5.C	IOC-1.A
16	A	5.D	IOC-2.A
17	B,C	3.B	AAP-3.A.a
18	A,D	1.D	AAP-2.L

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Computer Science Principles Exam page](#) on AP Central.



Scoring Guidelines

Video, Program Requirements, and Written Response 1

3 points

General Scoring Notes

- Responses should be evaluated solely on the rationale provided.
- Responses must demonstrate all criteria, including those within bulleted lists, in each row to earn the point for that row.
- Terms and phrases defined in the terminology list are italicized when they first appear in the scoring criteria.

Video, Program Requirements, and Written Response 1

3 points

Reporting Category	Scoring Criteria	Decision Rules
Course Project: Video (0-1 points)	The video demonstrates the running of the program including: <i>input</i> <i>program functionality</i> <i>output</i>	Consider only the video when scoring this point. Do NOT award a point if the following is true: The video does not show a demonstration of the program running (screenshots or storyboards are not acceptable and would not be credited).
Course Project: Program Requirements (0-1 points)	The program code includes: <i>A list</i> <i>A procedure</i> <i>A call to the procedure</i> <i>Selection</i> <i>Iteration</i>	Consider the Personalized Project Reference (or Program Code if necessary) when scoring this point. - If the program requirements do not appear in the Personalized Project Reference, consider the full program code file when scoring this point. - The procedure does not need to have a parameter to earn this point. - The code segments demonstrating selection and iteration do not need to appear in the same algorithm to earn this point. - The code segments demonstrating selection and iteration do not need to be contained in a procedure to earn this point. Do NOT award a point if any one or more of the following is true: - The list is a one-element list. - The use of the list is irrelevant or not used in the program. - The use of either the selection or the iteration is trivial (i.e., does not affect the outcome of the program).
Written Response 1: Program Design, Function, and Purpose (0-1 points)	The written response: - identifies an expected user of the program. - describes one way the program's design meets the needs of the identified user.	Consider Written Response 1 and the student's Program Code when scoring this point. - The response must relate the program design to the needs of the identified user. Do NOT award a point if the following is true: - The description of the design is implausible, inaccurate, or inconsistent with the program.

General Scoring Notes

- Responses should be evaluated solely on the rationale provided.
- Responses must demonstrate all criteria, including those within bulleted lists, in each row to earn the point for that row.
- Terms and phrases defined in the terminology list are italicized when they first appear in the scoring criteria.

Written Response 2

3 points

Reporting Category	Scoring Criteria	Decision Rules
Written Response 2(a): Algorithm Development (0-1 points)	The written response: - identifies the number of times the body of the iteration statement will execute. AND - describes a condition or error that would cause an infinite loop. OR - if no such condition or error exists, explains how the loop could be modified to cause an infinite loop.	Consider the Personalized Project Reference and Written Response 2(a) when scoring this point. - If multiple iteration statements are included in the Procedure section of the Personalized Project Reference, use the first iteration statement to determine whether the point is earned. - The iteration statement does not need to be contained in a procedure to earn this point. Do NOT award a point if the following is true: - The identified number of times the body of the iteration statement will execute does not match the code.
Written Response 2(b): Errors and Testing (0-1 points)	The written response: - describes a change to the procedure that will result in a run-time error. - explains why the change will result in a run-time error.	Consider the Personalized Project Reference and Written Response 2(b) when scoring this point. - If multiple procedures are included in part (i) of the Procedure section of the Personalized Project Reference, use the first procedure to determine whether the point is earned. Do NOT award a point if any one or more of the following is true: - A procedure is not identified in part (i) of the Procedure section of the Personalized Project Reference. - The response does not apply to the procedure in part (i) of the Procedure section of the Personalized Project Reference. - The response describes expected behavior that is implausible, inaccurate, or inconsistent with the program.
Written Response 2(c): Data and Procedural Abstraction (0-1 points)	The written response: explains in detailed steps an algorithm that uses <code>isEqual</code> to count the number of elements in the list that are equal to a certain value.	Consider the Personalized Project Reference and Written Response 2(c) when scoring this point. - If multiple lists are included in the List section of the Personalized Project Reference, use the first list to determine whether the point is earned. - The algorithm can be described in code, pseudocode, as a sequence of steps in English, or as a paragraph in English. - The algorithm must describe iterating over all elements of the list, calling <code>isEqual</code> on each list element and maintaining a count of the number of list elements equal to a certain value. - The algorithm must correctly determine the number of elements that are equal to a certain value. Some ways this can be determined is by storing the value in a variable, returning it, or displaying it to the user. Do NOT award a point if the following is true: - A list is not identified in the List section of the Personalized Project Reference. - The description of the algorithm is not detailed enough for someone else to write the code.